

10 M.Tech Project Topics in AI

NLP-Focused, with Computer Vision — 2026 Reference Guide

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Overview

This guide proposes **10 scoped, publishable M.Tech topics**: 7 in NLP (corrective RAG, hallucination detection, QLoRA for Indic LLMs, Hindi-English code-mixing, tool-use agents, small-model distillation, prompt-injection defense) and 3 in Computer Vision (deepfake/AI-image detection, small VLMs for document understanding, label-efficient medical segmentation).

Every topic is chosen for **2024-2026 research activity, public datasets, and single-GPU/Colab feasibility** — most are runnable on one 16-24 GB GPU using 7B-or-smaller models with QLoRA, frozen backbones, or API-based evaluation.

Where the feasible novelty is in 2026: the strongest wins for a single student lie in *evaluation, robustness, and efficiency* rather than training new foundation models. Hallucination detection, corrective RAG, code-mixed benchmarking, and parameter-efficient adaptation all offer publishable deltas on modest hardware. Indic/low-resource and code-mixed NLP remain notably under-served, with strong 2025-2026 dataset releases making them fertile ground.

TOPIC 1 — Corrective / Adaptive Retrieval-Augmented Generation for Domain QA

Area: NLP

Title

Confidence-Gated Corrective RAG: Reducing Unsupported Answers in Domain-Specific Question Answering

Problem Statement

Standard RAG degrades badly when retrieval returns irrelevant or partially relevant passages, causing the LLM to hallucinate or copy noise. Corrective/adaptive RAG methods that assess retrieval quality and trigger recovery actions (re-retrieve, web search, discard) are a hot 2024-2025 area but are under-tested on domain corpora and small open models. This matters now because enterprises increasingly deploy RAG on private documents where retrieval is imperfect.

Objective Goals

- Build a modular RAG pipeline with a lightweight retrieval-quality evaluator that classifies retrieved context as Correct / Ambiguous / Incorrect and triggers corrective actions.
- Implement and compare at least three strategies: vanilla RAG, Corrective RAG (CRAG-style), and Adaptive-RAG (complexity-routed retrieval).
- Evaluate on 2-3 QA benchmarks and demonstrate a measurable reduction in hallucination / unsupported-claim rate vs. vanilla RAG at equal or better answer accuracy.
- Quantify latency and cost overhead introduced by the corrective loop.

Methodology / Models

Retriever: BGE or E5 embeddings + FAISS. Generator: a 7-8B open model (Llama-3.1-8B-Instruct, Qwen2.5-7B, or Mistral-7B) via API or local 4-bit quantization. Train a small evaluator (fine-tuned DeBERTa / ModernBERT or a distilled scorer) for relevance grading. Add query-complexity routing (Adaptive-RAG) and self-correction prompts.

Datasets / Benchmarks

Natural Questions, HotpotQA (multi-hop), PopQA, TriviaQA. For domain specialization: PubMedQA or a legal/finance corpus. RAGTruth for span-level hallucination labels.

Evaluation Metrics

Exact Match / F1 (answer quality), retrieval recall@k, faithfulness / groundedness (LLM-as-judge or NLI entailment), unsupported-claim rate, end-to-end latency.

Key References

- Yan, Gu, Zhu, Ling. *Corrective Retrieval Augmented Generation (CRAG)*. arXiv 2024. <https://arxiv.org/abs/2401.15884>
- Jeong et al. *Adaptive-RAG: Learning to Adapt Retrieval-Augmented LLMs Through Question Complexity*. NAACL 2024. <https://arxiv.org/abs/2403.14403>
- Wang et al. *Astute RAG: Overcoming Imperfect Retrieval Augmentation and Knowledge Conflicts*. arXiv 2024. <https://arxiv.org/abs/2410.07176>
- Retrieval-Augmented Generation: A Comprehensive Survey of Architectures, Enhancements, and Robustness Frontiers*. arXiv 2025.

Feasibility

Very feasible. Retrieval plus API generation needs no GPU; local 7B inference runs on a single 16 GB GPU or Colab. The evaluator is a small model (under 1 GPU-hour to fine-tune). Main effort is engineering the pipeline and evaluation harness.

TOPIC 2 — Hallucination Detection in LLM / RAG Outputs

Area: NLP

Title

Span-Level Hallucination Detection for RAG: A Comparison of Black-Box and Fine-Tuned Detectors

Problem Statement

Even with RAG, LLMs produce unsupported or contradictory spans. Reliable, cheap, span-level hallucination detectors — ideally without access to model internals — are an active 2025–2026 problem. Recent work shows existing benchmarks understate hallucination prevalence and existing detectors remain weak on hard cases: on RAGTruth, a Llama-2-13B fine-tuned detector reaches only 52.7% overall span-level F1, underscoring the near-50% ceiling.

Objective Goals

1. Build a span-level hallucination detector for RAG outputs (input: query + retrieved context + generated answer; output: supported / contradicted / baseless spans).
2. Compare black-box methods (self-consistency, NLI entailment, LLM-as-judge) against lightweight fine-tuned detectors trained on RAGTruth.
3. Report detection F1 against annotated benchmarks and analyze failure modes by hallucination type.
4. Optionally add a mitigation step (abstain / regenerate) and measure the downstream faithfulness gain.

Methodology / Models

Fine-tune a modern encoder (DeBERTa-v3, ModernBERT) on RAGTruth span labels. Baselines: SelfCheckGPT-style sampling consistency, NLI models (e.g., DeBERTa-MNLI), and LLM-as-judge (GPT-4o-mini or an open 8B model). Explore uncertainty and semantic-entropy signals.

Datasets / Benchmarks

RAGTruth — nearly 18,000 naturally generated responses from diverse LLMs using RAG, manually annotated at case and word level (Niu et al., ACL 2024). Also HaluEval, FaithBench, and the HalluLens benchmark (ACL 2025).

Evaluation Metrics

Span- and response-level precision / recall / F1, AUROC, balanced accuracy, correlation with human labels.

Key References

- Niu et al. *RAGTruth: A Hallucination Corpus for Developing Trustworthy Retrieval-Augmented Language Models*. ACL 2024. <https://arxiv.org/abs/2401.00396>
- Bang et al. *HalluLens: LLM Hallucination Benchmark*. ACL 2025.
- Farquhar, Kossen, Kuhn, Gal. *Detecting Hallucinations in Large Language Models Using Semantic Entropy*. Nature 2024.
- Tamber et al. *Benchmarking LLM Faithfulness in RAG with Evolving Leaderboards (FaithJudge)*. arXiv 2025. <https://arxiv.org/abs/2505.04847>

Feasibility

Highly feasible. Encoder fine-tuning fits on one 16 GB GPU or Colab. Black-box baselines need only API calls. Data is public. Strong publishable angle in comparing detector families on a common benchmark.

TOPIC 3 — QLoRA Parameter-Efficient Fine-Tuning for a Low-Resource Indic Domain LLM

Area: NLP

Title

QLoRA vs. LoRA for Indic Domain Adaptation: A Memory-Accuracy Study on a 7B Open Model

Problem Statement

Full fine-tuning of LLMs is out of reach for most students, and Indic languages remain under-represented in open LLMs. QLoRA

(Dettmers et al., NeurIPS 2023) reduces memory usage enough to finetune a 65B parameter model on a single 48 GB GPU while preserving full 16-bit finetuning task performance; its Guanaco-65B model reached 99.3% of the performance level of ChatGPT while requiring only 24 hours of finetuning on a single GPU. Combining PEFT with an Indic language plus a domain (agriculture, health, legal QA) is a realistic and useful contribution.

Objective Goals

1. Fine-tune a 7-8B base model with QLoRA on a chosen Indic language and domain task (instruction-tuning or QA).
2. Compare LoRA vs. QLoRA vs. (where feasible) full fine-tuning on quality, trainable-parameter count, memory footprint, and wall-clock time.
3. Evaluate against the zero-shot base model and an existing Indic LLM (e.g., Airavata) on a public benchmark.
4. Ablate adapter rank and quantization effects.

Methodology / Models

Base: Llama-3.1-8B, Gemma-2-9B, Sarvam, or Qwen2.5-7B. PEFT via Hugging Face PEFT + bitsandbytes (4-bit NF4, double quantization, paged optimizers). Optionally compare DoRA / AdaLoRA variants.

Datasets / Benchmarks

IndicGenBench, IndicMMLU-Pro, Indic QA Benchmark, IndicGLUE. Domain data such as translated PubMedQA or IL-TUR for legal. Airavata's instruction data as reference.

Evaluation Metrics

Task accuracy / F1, ROUGE / chrF / BLEU for generation, trainable-parameter percentage, peak VRAM, training time, plus human evaluation on a sample.

Key References

- Dettmers et al. *QLoRA: Efficient Finetuning of Quantized LLMs*. NeurIPS 2023. <https://arxiv.org/abs/2305.14314>
- Gala et al. *Airavata: Introducing Hindi Instruction-tuned LLM*. arXiv 2024. <https://arxiv.org/abs/2401.15006>
- Singh et al. *IndicGenBench: A Multilingual Benchmark to Evaluate Generation Capabilities of LLMs on Indic Languages*. arXiv 2024. <https://arxiv.org/abs/2404.16816>
- *PEFT A2Z: Parameter-Efficient Fine-Tuning Survey for Large Language and Vision Models*. arXiv 2025. <https://arxiv.org/abs/2504.14117>

Feasibility

Designed for limited hardware — QLoRA fine-tunes a 7B model on a single 16-24 GB GPU (RTX 4090 / Colab A100). A full fine-tuning comparison may need a rented A100 for one run; otherwise skip it and cite published numbers.

TOPIC 4 — Hindi-English (Hinglish) Code-Mixed NLP with LLMs

Area: NLP

Title

Benchmarking and Adapting Open LLMs for Hindi-English Code-Mixed NLP Across Devanagari and Roman Scripts

Problem Statement

Hundreds of millions of users write code-mixed Hinglish, but LLMs degrade sharply on mixed and Romanized text, and benchmarks are fragmented. Recent 2025–2026 resources (COMI-LINGUA, code-switching surveys) make this a timely, high-impact, under-served area — especially Romanized script, where even Indic-tuned models struggle.

Objective Goals

1. Build or curate an evaluation suite for one or more Hinglish tasks (LID, POS, NER, sentiment, or MT) across both Devanagari and Roman scripts.
2. Benchmark open LLMs and Indic-tuned models zero-shot, few-shot, and after QLoRA fine-tuning.
3. Quantify degradation vs. monolingual baselines and correlate errors with code-mixing intensity (CMI).
4. Deliver a fine-tuned Hinglish model or adapter that improves over HingBERT / Airavata baselines on at least one task.

Methodology / Models

Baselines: HingBERT / HingRoBERTa (L3Cube), mBERT, XLM-R. LLMs: Llama-3.1-8B, Qwen2.5-7B, Airavata, Sarvam. Fine-tune with LoRA / QLoRA; use CMI and SymCoM to stratify test sets.

Datasets / Benchmarks

- **GLUECoS** — 11 datasets, 6 tasks (LID, POS, NER, sentiment, QA, NLI), En-Hi and En-Es. <https://arxiv.org/abs/2004.12376>
- **L3Cube-HingCorpus + HingBERT** — per Nayak & Joshi (WILDRE-6 @ LREC 2022), the first large-scale real Hindi-English code-mixed data in Roman script: 52.93M sentences and 1.04B tokens scraped from Twitter, plus HingBERT / HingRoBERTa / HingGPT

models. <https://arxiv.org/abs/2204.08398>

- **COMI-LINGUA** — 125k+ instances, 376k+ expert annotations, 5 tasks (token-level LID, matrix language ID, POS, NER, MT), both scripts. <https://arxiv.org/abs/2503.21670>
- **SemEval-2020 Task 9 (SentiMix)** — roughly 20k Hinglish tweets with word-level LID and sentence-level sentiment. <https://aclanthology.org/2020.semeval-1.100/>
- **DravidianCodeMix** — 60k+ annotated YouTube comments (Tamil / Kannada / Malayalam-English) for sentiment and offensive-language identification; useful for cross-family comparison. <https://arxiv.org/abs/2106.09460>

Evaluation Metrics

Weighted and macro F1 plus accuracy for classification; BLEU / chrF++ for MT; Code-Mixing Index (CMI), M-Index, and I-Index / switch-point fraction to characterize mixing; Fleiss' Kappa for any new annotation.

Key References

- Sheth, Beniwal, Singh. *COMI-LINGUA: Expert Annotated Large-Scale Dataset for Multitask NLP in Hindi-English Code-Mixing*. Findings of EMNLP 2025. <https://arxiv.org/abs/2503.21670>
- Sheth et al. *Beyond Monolingual Assumptions: A Survey of Code-Switched NLP in the Era of LLMs*. arXiv 2025 (ACL 2026, forthcoming). <https://arxiv.org/abs/2510.07037>
- Nayak, Joshi. *L3Cube-HingCorpus and HingBERT*. WILDRE-6 @ LREC 2022. <https://arxiv.org/abs/2204.08398>
- Khanuja et al. *GLUECoS: An Evaluation Benchmark for Code-Switched NLP*. ACL 2020. <https://arxiv.org/abs/2004.12376>

Feasibility

Very feasible. Encoder baselines and 7B QLoRA fine-tuning run on one GPU or Colab; LLM evaluation via API. All datasets are public. Strong path to a workshop or conference paper given how under-served the area is.

TOPIC 5 — Robustness and Failure Recovery in LLM Tool-Use Agents

Area: NLP

Title

Detect-and-Replan: Improving Failure Recovery in LLM Tool-Use Agents

Problem Statement

LLM agents that call tools and APIs fail silently on malformed calls, tool errors, and multi-step planning. 2025–2026 benchmarks (tau-bench, ToolSandbox, MCP-Bench) reveal large reliability gaps, especially for open models. Studying error detection and replanning is timely as agentic AI deploys widely.

Objective Goals

1. Build an agent (ReAct or native tool-calling) over a public tool benchmark and measure baseline success plus a failure taxonomy.
2. Implement at least two robustness mechanisms: self-verification of tool arguments, error-triggered replanning, and retry with reflection.
3. Evaluate improvement in task success and reduction of invalid tool calls on a benchmark.
4. Compare an open 7–8B model against a stronger API model to quantify the capability gap.

Methodology / Models

Frameworks: LangGraph or native function-calling. Models: Qwen2.5-7B-Instruct, Llama-3.1-8B, plus GPT-4o-mini as a ceiling. Add a verifier module and a replanning loop.

Datasets / Benchmarks

tau-bench (ICLR 2025), ToolSandbox (arXiv 2408.04682), BFCL (Berkeley Function-Calling Leaderboard), API-Bank, MCP-Bench (arXiv 2508.20453).

Evaluation Metrics

Task success rate, tool-call validity / accuracy, number of steps, recovery rate after induced errors, cost and latency.

Key References

- Yao, Shinn, Razavi, Narasimhan. *tau-bench: A Benchmark for Tool-Agent-User Interaction in Real-World Domains*. ICLR 2025.
- Lu et al. *ToolSandbox: A Stateful, Conversational, Interactive Evaluation Benchmark for LLM Tool Use*. NAACL Findings 2025. <https://arxiv.org/abs/2408.04682>
- Wang et al. *MCP-Bench: Benchmarking Tool-Using LLM Agents with Complex Real-World Tasks via MCP Servers*. arXiv 2025. <https://arxiv.org/abs/2508.20453>
- Patil, Zhang, Wang, Gonzalez. *Gorilla: Large Language Model Connected with Massive APIs*. arXiv 2023. <https://arxiv.org/abs/2305.15334>

Feasibility

Feasible with API models (no GPU) or one 16 GB GPU for a local 7B model. Main cost is engineering the harness and, if using an API, inference credits. Simulated tools avoid real API costs.

TOPIC 6 — Small Language Model Reasoning via Knowledge Distillation

Area: NLP

Title

Distilling Chain-of-Thought into Sub-3B Models: A Comparison of Distillation Objectives

Problem Statement

Large reasoning models are costly; distilling their chain-of-thought rationales into sub-3B models that run on edge devices or Colab is an active efficiency frontier. The gains can be dramatic — Hsieh et al.'s "Distilling Step-by-Step" (ACL 2023 Findings) shows a finetuned 770M T5 model outperforming the few-shot prompted 540B PaLM model using only 80% of available data, a greater-than-700x model-size reduction — making this a publishable, compute-light direction.

Objective Goals

1. Distill reasoning (CoT rationales) from a strong teacher into a small student (3B parameters or fewer) on a reasoning task family.
2. Compare distillation strategies: standard SFT on teacher rationales (distilling step-by-step), sequence-level KD, and a program-aided or on-policy variant.
3. Show the small student outperforms a same-size SFT baseline and approaches the teacher on a benchmark at a fraction of parameters and FLOPs.
4. Analyze the accuracy vs. size trade-off (capacity gap).

Methodology / Models

Teacher: an API model or a 32B+ open reasoning model. Student: Qwen2.5-1.5B/3B, Llama-3.2-1B/3B, or Gemma-2-2B. Generate rationales, filter them, fine-tune with LoRA, and try a DistiLLM-2-style contrastive objective.

Datasets / Benchmarks

GSM8K, MATH (subset), StrategyQA, CommonsenseQA, ARC, BBH.

Evaluation Metrics

Task accuracy, parameters / FLOPs, tokens per second, accuracy-per-parameter, robustness on held-out reasoning types.

Key References

- Hsieh et al. *Distilling Step-by-Step! Outperforming Larger LMs with Less Training Data and Smaller Model Sizes*. ACL 2023 Findings. <https://arxiv.org/abs/2305.02301>
- Gu et al. *MiniPLM: Knowledge Distillation for Pre-Training Language Models*. arXiv 2024/2025. <https://arxiv.org/abs/2410.17215>
- Ko et al. *DistiLLM-2: A Contrastive Approach Boosts the Distillation of LLMs*. arXiv 2025. <https://arxiv.org/abs/2503.07067>
- Zhu et al. *PaD: Program-aided Distillation*. arXiv 2023. <https://arxiv.org/abs/2305.13888>

Feasibility

Student fine-tuning (3B or smaller, LoRA) fits one 16–24 GB GPU. Teacher rationales come via API (budget for generation) or a one-off rented GPU. All data is public.

TOPIC 7 — Prompt-Injection Detection and Guardrails for LLM Applications

Area: NLP

Title

Building and Red-Teaming a Prompt-Injection Guardrail for RAG / Agent Pipelines

Problem Statement

Prompt injection (direct and indirect) is OWASP's number-one LLM risk, and RAG / agent apps are especially exposed via retrieved and tool-returned content. 2025–2026 work shows existing guardrails are bypassable, so building and stress-testing detectors is timely and practically valuable.

Objective Goals

1. Build a prompt-injection detection classifier and integrate it as a guardrail in a RAG or agent pipeline.
2. Evaluate detection on public benchmarks covering both direct and indirect injection.

- 3. Red-team the guardrail with evasion techniques (character injection, obfuscation) and report the bypass rate.
- 4. Compare a fine-tuned detector against off-the-shelf guardrails (Prompt Guard, open classifiers).

Methodology / Models

Fine-tune ModernBERT or DeBERTa on injection datasets. Baselines: Llama Prompt Guard, structured-query defenses (StruQ). Test indirect injection via poisoned retrieved documents.

Datasets / Benchmarks

BIPIA (indirect injection benchmark), formalized prompt-injection attack/defense suites, JailbreakBench, and SoK / PIArena resources.

Evaluation Metrics

Detection precision / recall / F1, AUROC, attack success rate (ASR) before and after the guardrail, false-positive rate on benign inputs, latency.

Key References

- Chen, Piet, Sitawarin, Wagner. *StruQ: Defending Against Prompt Injection with Structured Queries*. arXiv 2024. <https://arxiv.org/abs/2402.06363>
- *Bypassing Prompt Injection and Jailbreak Detection in LLM Guardrails*. arXiv 2025. <https://arxiv.org/abs/2504.11168>
- *SoK: Systematizing LLM Prompt Security: Taxonomies, Datasets, and Unified Evaluation of Attacks and Defenses*. arXiv 2025. <https://arxiv.org/abs/2510.15476>
- OWASP GenAI Security Project. *LLM01:2025 Prompt Injection*. <https://genai.owasp.org/llmrisk/llm01-prompt-injection/>

Feasibility

Encoder fine-tuning and evaluation run on one GPU or Colab; attacks and evaluation via API. Public datasets. Clear, self-contained, and publishable.

TOPIC 8 — Generalizable Deepfake / AI-Generated Image Detection

Area: Computer Vision

Title

Frozen-Backbone, Parameter-Efficient Deepfake Detection that Generalizes Across Generators

Problem Statement

Detectors trained on one generator fail to generalize to unseen manipulations and generators — the core open problem. 2025 work shows parameter-efficient adaptation of frozen vision foundation models (CLIP, DINOv2) generalizes better than complex architectures: Yermakov et al.'s GenD fine-tunes only the Layer Normalization parameters (0.03% of the total) and achieves state-of-the-art performance, outperforming more complex recent approaches in average cross-dataset AUROC across 14 benchmarks spanning 2019–2025. Strong results are therefore achievable on modest compute.

Objective Goals

- 1. Build a cross-dataset deepfake / AI-image detector using a frozen foundation backbone with parameter-efficient adaptation (LayerNorm or LoRA tuning).
- 2. Train on one dataset and evaluate cross-dataset (train on FF++ or GenImage; test on Celeb-DF, DFDC, DF40, and recent generators).
- 3. Beat a from-scratch CNN baseline and match or approach recent methods in average cross-dataset AUROC.
- 4. Analyze what improves generalization: paired real/fake sampling, augmentation, feature normalization.

Methodology / Models

Frozen CLIP-ViT or DINOv2 / DINOv3 encoder; tune only LayerNorm plus a linear or metric head (L2-normalized hyperspherical features). Baselines: Xception, standard CLIP linear-probe.

Datasets / Benchmarks

FaceForensics++, Celeb-DF v2, DFDC, DeepfakeBench, DF40 (next-generation), GenImage (AI-generated images).

Evaluation Metrics

AUROC and accuracy (in-domain and cross-dataset), average cross-dataset AUROC, average precision, robustness to compression.

Key References

- Yermakov et al. *Deepfake Detection that Generalizes Across Benchmarks (GenD)*. arXiv 2025. <https://arxiv.org/abs/2508.06248>
- Yan et al. *DF40: Toward Next-Generation Deepfake Detection*. NeurIPS 2024. <https://arxiv.org/abs/2406.13495>
- Yan et al. *DeepfakeBench: A Comprehensive Benchmark of Deepfake Detection*. NeurIPS 2023. <https://arxiv.org/abs/2307.01426>
- *Is AI-Generated Image Detection a Solved Problem?* arXiv 2025. <https://arxiv.org/abs/2505.12335>

Feasibility

Excellent. A frozen backbone plus a tiny trainable head trains on one 12-16 GB GPU or Colab, updating only 0.03-1% of parameters. Datasets are public, though some require access forms. Strong publishable generalization story.

TOPIC 9 — Small Vision-Language Model for Document Understanding / OCR

Area: Computer Vision

Title

Compact VLMs for Multi-Page Document Understanding: An Accuracy-Efficiency Study

Problem Statement

Multimodal document understanding (DocVQA, OCR-free extraction) is dominated by large VLMs. Compact VLMs that run on limited hardware for long and multi-page documents are an active 2025-2026 need for on-premise and low-cost deployment.

Objective Goals

- 1. Fine-tune or adapt a small VLM (3B parameters or fewer) for a document task — DocVQA, key-information extraction, or OCR-to-markdown — using LoRA.
- 2. Benchmark against OCRBench v2 and DocVQA, and compare against a larger VLM ceiling.
- 3. Improve efficiency (token count, memory) for multi-page inputs and quantify the accuracy / efficiency trade-off.
- 4. Optionally target a specific domain: invoices, forms, or Indic-script documents.

Methodology / Models

Small VLMs: Qwen2.5-VL-3B, InternVL2-2B/2.5, SmolVLM, or mPLUG-DocOwl2. LoRA fine-tuning. Compare OCR-pipeline against OCR-free approaches.

Datasets / Benchmarks

DocVQA, InfographicVQA, OCRBench and OCRBench v2, OmniDocBench, DUDE, plus olmOCR data for OCR-to-text.

Evaluation Metrics

ANLS (DocVQA), F1 / accuracy (key-information extraction), edit distance and CER/WER (OCR), OCRBench score, token count and VRAM.

Key References

- DocSLM: A Small Vision-Language Model for Long Multimodal Document Understanding. arXiv 2025. <https://arxiv.org/abs/2511.11313>
- Fu et al. OCRBench v2: An Improved Benchmark for Evaluating Large Multimodal Models on Visual Text Localization and Reasoning. arXiv 2025. <https://arxiv.org/abs/2501.00321>
- Poznanski et al. olmOCR: Unlocking Trillions of Tokens in PDFs with Vision Language Models. arXiv 2025. <https://arxiv.org/abs/2502.18443>
- Hu et al. mPLUG-DocOwl2: High-Resolution Compressing for OCR-Free Multi-Page Document Understanding. arXiv 2024. <https://arxiv.org/abs/2409.03420>

Feasibility

Feasible with a 3B-or-smaller VLM plus LoRA on a single 24 GB GPU (or Colab A100). High-resolution and multi-page training is the main memory pressure — mitigate with image tiling / compression and gradient checkpointing.

TOPIC 10 — Label-Efficient Medical Image Segmentation

Area: Computer Vision

Title

Matching Fully-Supervised Segmentation with Few Labels: A Semi- and Self-Supervised Study on CT/MRI

Problem Statement

Expert pixel-level labels for CT and MRI are scarce and expensive. Self-supervised pretraining and semi-supervised learning that match fully-supervised quality with only a few labels form a durable, clinically important research area with strong 2025 activity.

Objective Goals

- 1. Implement a semi- or self-supervised segmentation pipeline (contrastive or SSL pretraining followed by fine-tuning, or consistency /

- pseudo-label semi-supervision).
- Evaluate at low label budgets (5%, 10%, 20% labeled) against a fully-supervised upper bound and a supervised-only lower bound.
 - Show Dice improvement over the supervised baseline at each budget on at least two public datasets.
 - Analyze robustness to noisy pseudo-labels and domain shift across scanners.

Methodology / Models

Backbone: U-Net, nnU-Net, UNETR, or Swin-UNETR. Self-supervision: contrastive (global plus local) or masked autoencoding. Semi-supervision: mean-teacher, cross / dual-network consistency, high-confidence pseudo-labels.

Datasets / Benchmarks

ACDC (cardiac MRI), Synapse multi-organ CT, LA (left atrium), BraTS, MSD tasks, TotalSegmentator subsets.

Evaluation Metrics

Dice similarity coefficient (DSC), Hausdorff distance (HD95), average surface distance (ASD), per-label-budget performance curves.

Key References

- Domain and Task-Focused Example Selection for Data-Efficient Contrastive Medical Image Segmentation*. arXiv 2025. <https://arxiv.org/abs/2505.19208>
- Semi-Supervised Medical Image Segmentation via Dual Networks*. arXiv 2025. <https://arxiv.org/abs/2505.17690>
- You et al. *Mine Your Own Anatomy (MONA): Revisiting Medical Image Segmentation with Extremely Limited Labels*. arXiv 2022. <https://arxiv.org/abs/2209.13476>
- SGTC: Semantic-Guided Triplet Co-training for Sparsely Annotated Semi-Supervised Medical Image Segmentation*. arXiv 2024. <https://arxiv.org/abs/2412.15526>

Feasibility

2D segmentation runs on one 12–16 GB GPU or Colab. 3D volumetric training is heavier — use patch-based training, smaller volumes (ACDC, LA), and mixed precision. Datasets are public but require registration.

Recommendations

Choosing by available hardware

If you want the safest, fastest path to a publishable result on a laptop-class GPU: choose Topic 2 (hallucination detection), Topic 4 (Hinglish code-mixing), Topic 7 (prompt-injection defense), or Topic 8 (deepfake detection). All use frozen or encoder-scale models or API calls, have ready public data, and offer clear metric deltas.

If you have a single 24 GB GPU (RTX 3090/4090) or Colab Pro+ A100: Topics 1, 3, 5, 6, and 9 all become comfortable (7B QLoRA or small-VLM LoRA).

Staged 8–12 month plan

Phase	Months	Focus
1	1–2	Reproduce a baseline from the cited paper; lock the evaluation harness
2	3–5	Implement your proposed method or adapter
3	6–8	Run ablations plus cross-dataset / cross-lingual generalization
4	9–12	Write up; target a workshop (ACL/EMNLP/NAACL, CVPR/ICCV) then a main-conference short paper

Decision thresholds

If your method does not beat the baseline by a clear margin (more than 1–2 points of F1 / Dice / AUROC with statistical significance) by month 6, **pivot to a robustness / generalization or efficiency framing** — these publish even with modest accuracy gains.

If compute becomes a blocker, switch training-heavy topics (3, 6, 9) to evaluation-heavy ones (2, 4, 5, 7).

De-risk your references: prefer datasets you can download today. For gated medical and deepfake sets, submit access requests in month 1.

Caveats

- Some cited papers are very recent preprints (2025–2026) and a few are listed as forthcoming or accepted (e.g., the code-switching survey at ACL 2026, HalluLens at ACL 2025). Confirm the final venue and version before formal citation, and prefer peer-reviewed versions where available.
- Benchmark leaderboards move fast; specific state-of-the-art numbers will shift during your project. Always re-run baselines yourself rather than quoting paper numbers.

- Dataset sizes for COMI-LINGUA differ across versions (approximately 100k vs. 125k vs. 181k) — cite the exact release you use. Note that CS-Sum (arXiv 2505.13559) covers Mandarin, Tamil, and Malay-English, **not** Hindi-English, so do not treat it as a Hinglish resource.
- LLM-as-judge metrics (used in the RAG, hallucination, and agent topics) have known biases. Pair them with human evaluation on a sample for credibility.
- Medical and deepfake datasets carry ethical and usage restrictions. Obtain approvals, and never deploy detectors as sole decision-makers.